

Review of Chapter 2.

§2.1.

1. General population Model.

$$\frac{dP}{dt} = (\beta - \delta) P.$$

where $\beta = \beta(t)$, $\delta = \delta(t)$, $P = P(t)$.

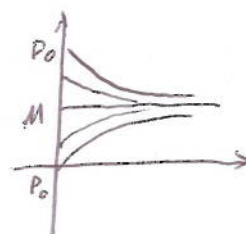
2. Logistic equation (Bounded Population)

$$\frac{dP}{dt} = aP - bP^2.$$

where a, b are both positive constants

Initial value problem

$$\begin{cases} \frac{dP}{dt} = kP(M-P) \\ P(0) = P_0 \end{cases} \rightarrow a = kM, b = k.$$

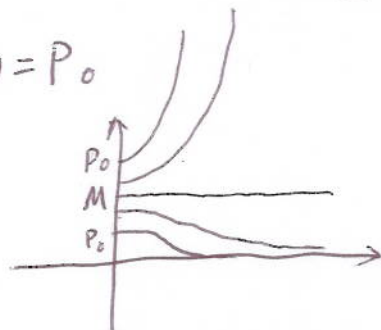


$$\Rightarrow P(t) = \frac{MP_0}{P_0 + (M - P_0)e^{-kMt}}$$

$$\lim_{t \rightarrow +\infty} P(t) = \frac{MP_0}{P_0 + 0} = M. \text{ — limiting population.}$$

3. Domsday vs. Extinction.

$$\begin{cases} \frac{dP}{dt} = kP(P-M) \\ P(0) = P_0 \end{cases} \Rightarrow P(t) = \frac{MP_0}{P_0 + (M - P_0)e^{kMt}}.$$



M — threshold population.

i) $P_0 < M$

$$\lim_{t \rightarrow +\infty} P(t) = 0$$

ii) $P_0 > M$

$$\lim_{t \rightarrow +\infty} P(t) = +\infty$$

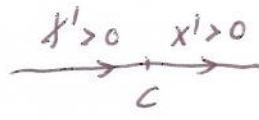
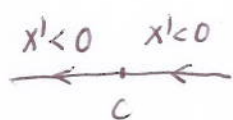
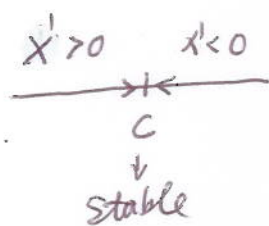
§22.

Autonomous first-order differential equation

$$\frac{dx}{dt} = f(x)$$

Defn: The solution of $f(x)=0$ are called critical points. If $x=C$ is a critical point, then $x(t) \equiv C$ is called an equilibrium solution of $\frac{dx}{dt} = f(x)$.

Defn: A critical point $x=C$ of $\frac{dx}{dt} = f(x)$ is said to be stable provided that if the initial value x_0 is sufficiently close to C , the $x(t)$ remains close to C for all $t > 0$.



unstable

§23. Motion under gravitational force and resistance force.

$$m \frac{dv}{dt} = F_G + F_R$$

$\downarrow$ Gravitational force. $\hookrightarrow$ resistance force.

$$F_R = -kV^P$$

where $1 \leq P \leq 2$, k a constant.

$$\Rightarrow m \frac{dv}{dt} = -mg - kV^P$$

Case I: $\begin{cases} \frac{P=1}{m \frac{dv}{dt} = -kV - mg} \\ V(0) = V_0 \end{cases} \Rightarrow V(t) = (V_0 + \frac{g}{\rho})e^{-\rho t} - \frac{g}{\rho}$

where $\rho = \frac{k}{m}$.

$\lim_{t \rightarrow \infty} V(t) = -\frac{g}{\rho} \rightarrow$ terminal speed $|V_0| = \frac{g}{\rho} = \frac{mg}{k}$.

Case II: $P=2$.

upward motion: $\frac{dv}{dt} = -g - \rho v^2$ (Since $v > 0$)

$$v(t) = \sqrt{\frac{g}{\rho}} \tanh\left(\sqrt{\frac{\rho}{g}} v_0 - \sqrt{\rho g} t\right)$$

downward motion: $\frac{dv}{dt} = -g + \rho v^2$ (Since $v < 0$)

$$v(t) = \sqrt{\frac{g}{\rho}} \tanh(c - t\sqrt{\rho g}) \quad \text{where } c = \tanh^{-1}(v_0 \sqrt{\frac{\rho}{g}}).$$

§24.

Consider:

$$\frac{dy}{dx} = f(x, y), \quad y(x_0) = y_0$$

Euler's Algorithm

step size: h .

Iterative formula:
$$\begin{cases} x_{n+1} = x_n + h \\ y_{n+1} = y_n + h \cdot f(x_n, y_n) \end{cases} \quad (n = 0, 1, 2, \dots, k)$$

$$\Rightarrow y_n \approx y(x_n)$$

In fact

$$|y_n - y(x_n)| \leq Ch \quad \text{for each } n = 1, 2, 3, \dots, k.$$